

SLIMANE DRIDI, PHD

Data Scientist | Mathematical & Statistical Modelling | Python

slimane.dridi@gmail.com | 06 67 86 42 01 | Toulouse, France | Full driving licence

slimaned.github.io

Skills

Statistical inference | Stochastic processes | Simulation | Time series | Experimental design | Model validation | Game theory

Python | NumPy | SciPy | pandas | R | Git | SQL (basics) | C++ | Mathematica

Languages

French: Native

English: Fluent

Modern Standard Arabic: Intermediate

Spanish: Basic conversational proficiency

Selected work

- Developed pyNamo-EGT, a Python package for equilibrium analysis and visualisation of evolutionary game dynamics (GitHub).
- Built a validated pressing-aggression index from event and player-tracking data, testing an evolutionary game theory model against 1,500+ matches across Europe's top leagues (GitHub).
- Co-authored seven scientific articles, presented at ten conferences and seminars, and reviewed manuscripts for ten international journals (Portfolio).

Profile

Data scientist and mathematical modeller with 10+ years of experience building, simulating and validating quantitative models in France, Switzerland and the United States. My research focused on how agents learn and respond to incentives (reinforcement learning, evolutionary game theory), and I apply the same tools to real data: statistical modelling, simulation and model validation in Python. I want to bring this modelling rigour to real-world decision and performance problems, in industry and in sport.

Professional Experience

- Independent Quantitative Research, Financial Markets | Self-employed

  09/2023–12/2025 | Toulouse, France

  - Built Python tools for market data analysis, statistical inference and strategy evaluation.
  - Designed and backtested systematic trading strategies (pairs mean reversion) on historical data, with a focus on robustness and out-of-sample performance.
  - Developed an algorithmic trading system and assessed its limits under real-world conditions.

- Mathematics Instructor, Nursing Entrance Examination | French Red Cross

  12/2024–01/2026 | Toulouse, France

- Mathematics, Statistics & Programming Tutor | Acadomia

  09/2023–06/2026 | Toulouse, France

- Researcher | Toulouse School of Economics

  04/2021–08/2021 | Toulouse, France

- Research Fellow | Institute for Advanced Study in Toulouse

  09/2017–07/2020 | Toulouse, France

  - Developed and analysed stochastic models of reinforcement learning and cooperation in repeated games; checked analytical predictions against simulations.

- Postdoctoral Researcher | University of Pennsylvania

  08/2015–08/2017 | Philadelphia, PA, USA

  - Modelled the evolution of learning in mathematical biology.

- Postdoctoral Researcher | University of Lausanne

  03/2014–02/2015 | Lausanne, Switzerland

Education

- PhD in Ecology and Evolution | University of Lausanne

  2009–2013 | Lausanne, Switzerland

- Master's Degree in Cognitive Science | University of Grenoble

  09/2007–09/2009 | Grenoble, France

- Bachelor's Degree in Psychology | University of Grenoble

Interests

- Sports (daily practice):  , 
- Mountain hiking.
- Volunteering: outreach work supporting people experiencing homelessness.

09/2004–09/2007 Grenoble, France

Teaching Experience

• Research Project Supervision | Toulouse School of Economics, University of Pennsylvania, University of Lausanne

2013–2020

- Supervised three Master's projects and one undergraduate research project.

• Teaching Assistant | Toulouse School of Economics, University of Lausanne

2013–2018

- Linear algebra, population genetics, and experimental design.